



WMSU

Foundations of HCI: The Computer



Learning Outcomes

- Demonstrate understanding on the properties of the devices with which a system is built.
- Classify different physical controls and displays
- Examine a range of input and output devices in order to understand how these influence interaction.
- Design a device and analyze how its properties or attributes influence interaction



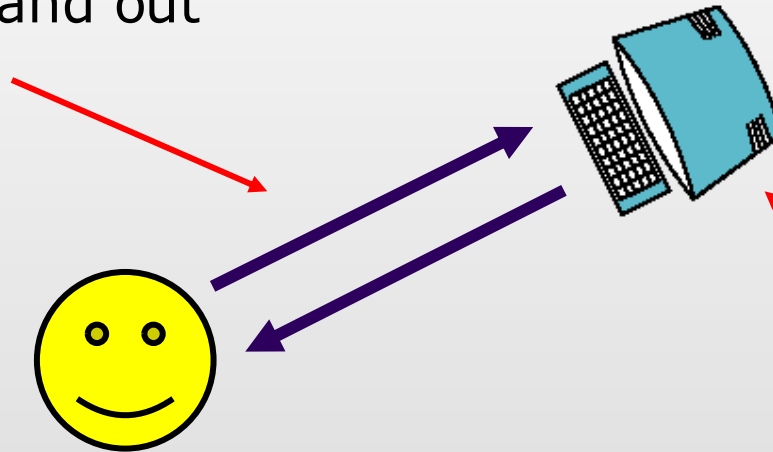
The Computer

- a computer system is made up of various elements
- each of these elements affects the interaction
 - input devices – text entry and pointing
 - output devices – screen (small & large), digital paper
 - virtual reality – special interaction and display devices
 - physical interaction – e.g. sound, haptic, bio-sensing
 - paper – as output (print) and input (scan)
 - memory – RAM & permanent media, capacity & access
 - processing – speed of processing, networks

Interacting with computers

to understand human–*computer* interaction
... need to understand computers!

what goes in and out
devices, paper,
sensors, etc.



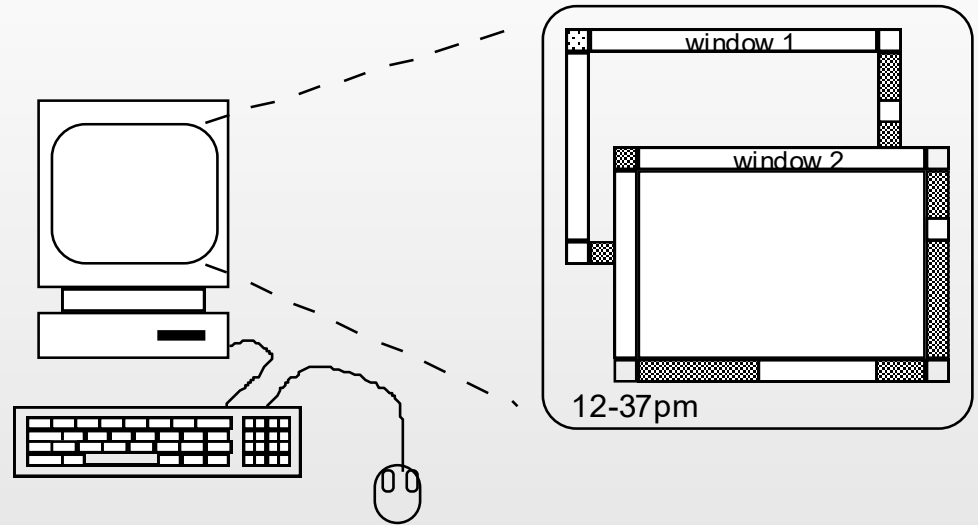
what can it do?
memory, processing,
networks

A 'typical' computer system



- screen, or monitor, on which there are windows
- keyboard
- mouse/trackpad

- variations
 - desktop
 - laptop
 - PDA



the devices dictate the styles of interaction that the system supports

If we use different devices, then the interface will support a different style of interaction



How many computers ...

in your house?

- PC
- TV
- microwave, cooker, washing machine
- central heating
- security system

can you think of more?

in your pockets?

- PDA
- phone, camera
- smart card, card with magnetic strip?
- electronic car key
- USB memory

try your pockets and bags



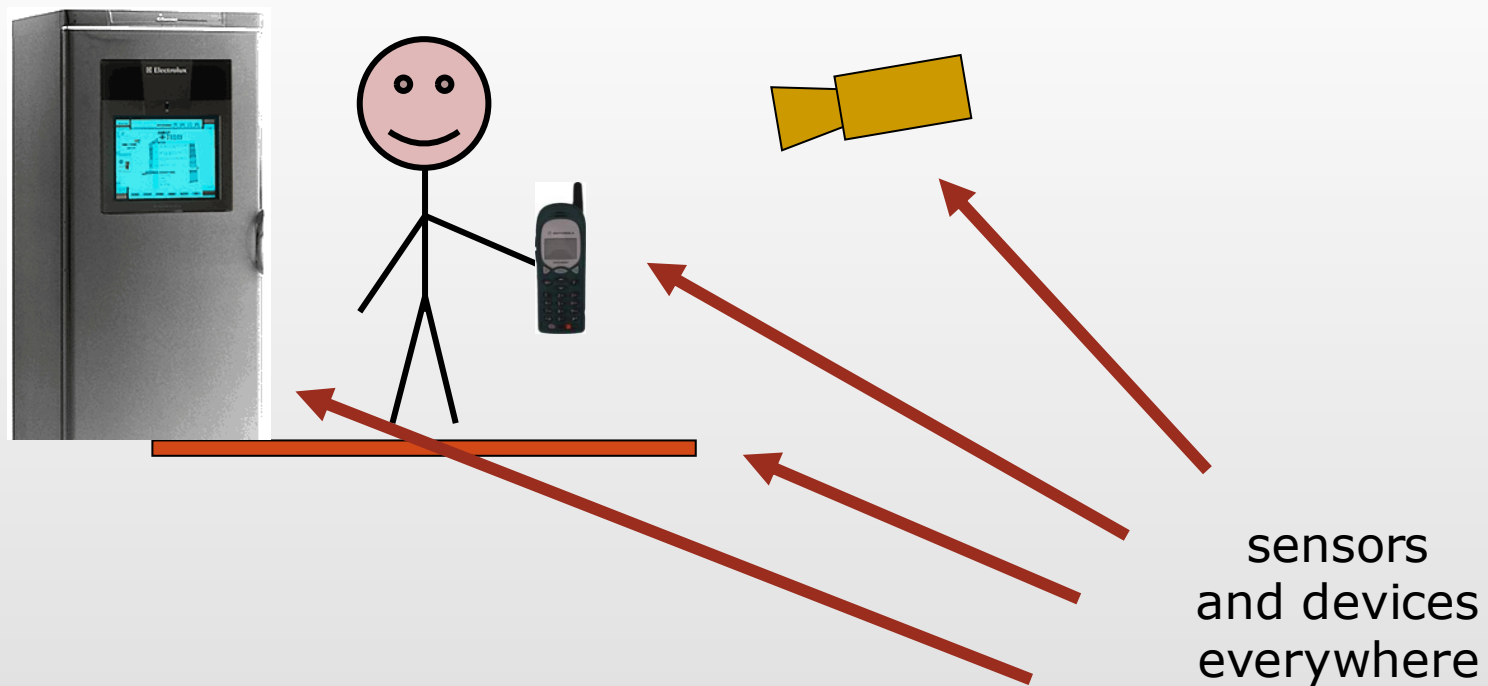
Interactivity?

- In the early days of computing, information was entered into the computer in a large mass – batch data entry.
 - using punched card stacks or large data files prepared
 - the interactions take place over hours or days

Now most computing is interactive

- rapid feedback
- the user in control (most of the time)
- doing rather than thinking ...

Richer interaction





TEXT ENTRY DEVICES



Keyboards

- Most common text input device
- Allows rapid entry of text by experienced users
- Key press closes connection, causing a character code to be sent
- Usually connected by cable, but can be wireless



layout – QWERTY

- Standardized layout
but ...
 - non-alphanumeric keys are placed differently
 - accented symbols needed for different scripts
 - minor differences between UK and USA keyboards
- QWERTY arrangement not optimal for typing
 - layout to prevent typewriters jamming!
- Alternative designs allow faster typing but large social base of QWERTY typists produces reluctance to change.

QWERTY keyboard



Alternative Keyboard Layouts

Alphabetic Keyboards : Ease of Learning

- keys arranged in alphabetic order
- not faster for trained typists
- not faster for beginners either!



DVORAK keyboard: Ergonomics of use

- common letters under dominant fingers
- biased towards right-handed people, in that 56% of keystrokes are made with the right hand.
- common combinations of letters alternate between hands
- 10-15% improvement in speed and reduction in fatigue

~	1	2	3	4	5	6	7	8	9	0	[	]	delete
tab	'	,	.	p	y	f	g	c	r	l	/	=	\
caps lock	a	o	e	u	i	d	h	t	n	s	-	return	
shift		;	q	j	k	x	b	m	w	v	z	shift	

~	!	@	#	\$	%	^	&	*	(	)	{	}	delete
tab	"	<	>	P	Y	F	G	C	R	L	?	+	
caps lock	A	O	E	U	I	D	H	T	N	S	_	return	
shift		:	Q	J	K	X	B	M	W	V	Z	shift	

Special Keyboards

- designs to reduce fatigue for RSI
- for one handed use
 - e.g. the Maltron left-handed keyboard



Chord keyboards

- only a few keys - four or 5
- letters typed as combination of keypresses
- compact size
 - ideal for portable applications
- short learning time
 - keypresses reflect letter shape
- fast
 - once you have trained



Phone Pad and T9 Entry

- use numeric keys with multiple presses

2 – a b c 6 - m n o

3 - d e f 7 - p q r s

4 - g h i 8 - t u v

5 - j k l 9 - w x y z

hello = 4433555[pause]555666

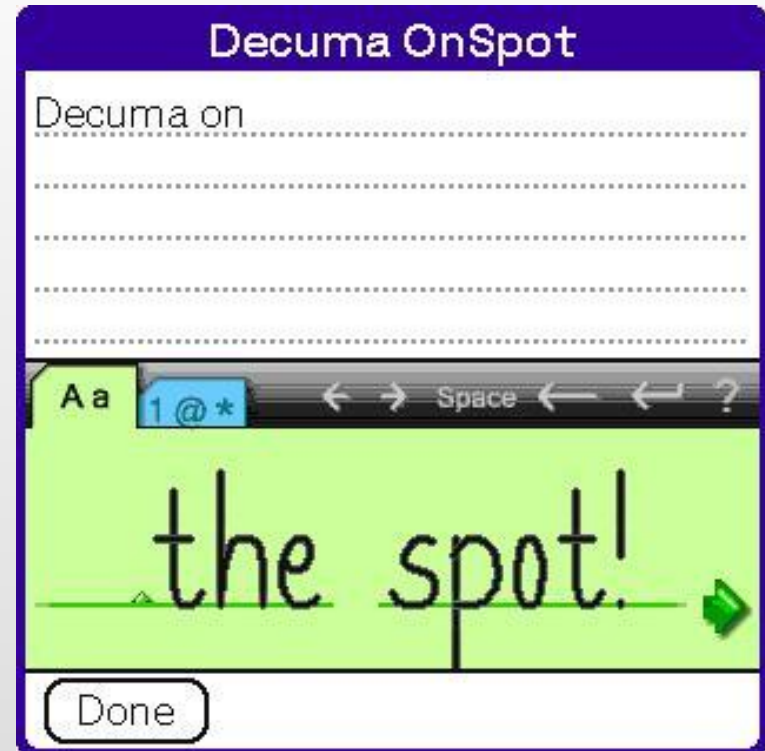
surprisingly fast!

- T9 predictive entry
 - type as if single key for each letter
 - use dictionary to ‘guess’ the right word
 - hello = 43556 ...
 - but 26 -> menu ‘am’ or ‘an’



Handwriting Recognition

- Text can be input into the computer, using a pen and a digesting tablet
 - natural interaction





Speech Recognition

- Improving rapidly
- Most successful when:
 - single user – initial training and learns peculiarities
 - limited vocabulary systems
- Problems with
 - external noise interfering
 - imprecision of pronunciation
 - large vocabularies
 - different speakers

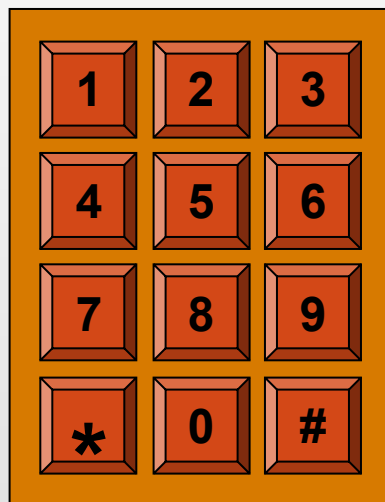


Numeric Keypads

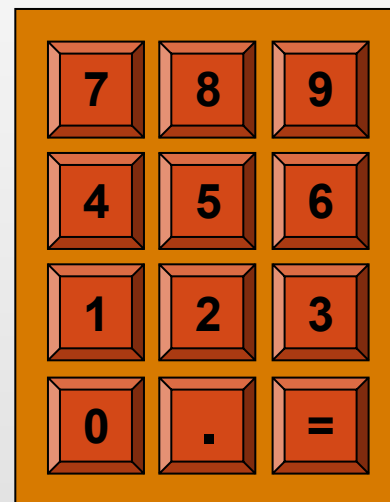
- for entering numbers quickly:
 - calculator, PC keyboard
- for telephones

not the same!!

ATM like phone



telephone



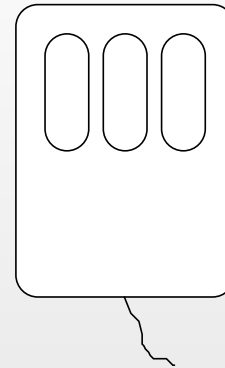
calculator



POSITIONING, POINTING, AND DRAWING

the Mouse

- Handheld pointing device
 - very common
 - easy to use
- Two characteristics
 - planar movement
 - buttons



(usually from 1 to 3 buttons on top, used for making a selection, indicating an option, or to initiate drawing etc.)



How does it work?

Two methods for detecting motion

- Mechanical
 - Ball on underside of mouse turns as mouse is moved
 - Rotates orthogonal potentiometers
 - Can be used on almost any flat surface
- Optical
 - light emitting diode on underside of mouse
 - may use special grid-like pad or just on desk
 - less susceptible to dust and dirt
 - detects fluctuating alterations in reflected light intensity to calculate relative motion in (x, z) plane



Even by foot ...

- some experiments with the *footmouse*
 - controlling mouse movement with feet ...
 - not very common :-)
- but foot controls are common elsewhere:
 - car pedals
 - sewing machine speed control
 - organ and piano pedals

Touchpad

- small touch sensitive tablets
- 'stroke' to move mouse pointer
- used mainly in laptop computers
- good 'acceleration' settings important
 - fast stroke
 - lots of pixels per inch moved
 - initial movement to the target
 - slow stroke
 - less pixels per inch
 - for accurate positioning



Trackball and thumbwheels

Trackball

- ball is rotated inside static housing
 - like an upside down mouse!
- relative motion moves cursor
- indirect device, fairly accurate
- separate buttons for picking
- very fast for gaming
- used in some portable and notebook computers.



Thumbwheels ...

- Thumbwheels are different in that they have two orthogonal dials to control the cursor position.
- Such a device is very cheap, but slow, and it is difficult to manipulate the cursor in any way other than horizontally or vertically.

Joystick and keyboard nipple

Joystick

- indirect
pressure of stick = velocity of movement
- buttons for selection
on top or on front like a trigger
- often used for computer games
aircraft controls and 3D navigation



Keyboard nipple

- for laptop computers
- miniature joystick in the middle of the keyboard

Touch-sensitive screen

- Detect the presence of finger or stylus on the screen.
 - works by interrupting matrix of light beams, capacitance changes or ultrasonic reflections
 - *direct* pointing device



Advantages:

- fast, and requires no specialised pointer
- good for menu selection
- suitable for use in hostile environment: clean and safe from damage.

Disadvantages:

- finger can mark screen
- imprecise
 - difficult to select small regions or perform accurate drawing
- lifting arm can be tiring



Stylus and light pen

Stylus

- small pen-like pointer to draw directly on screen
- may use touch sensitive surface or magnetic detection
- used in PDA, tablets PCs and drawing tables

Light Pen

- now rarely used
- uses light from screen to detect location

BOTH ...

- very direct and obvious to use
- but can obscure screen

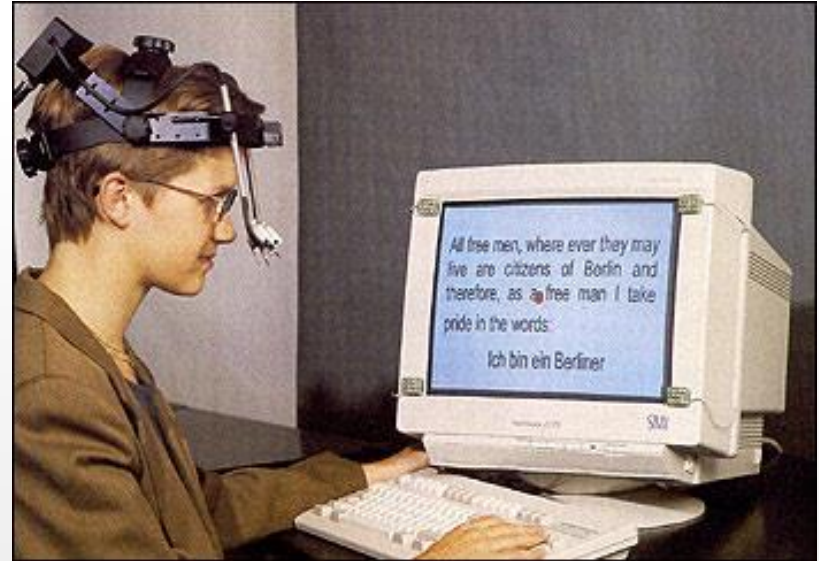
Digitizing tablet

- Mouse like-device with cross hairs
- used on special surface
 - rather like stylus
- very accurate
 - used for digitizing maps



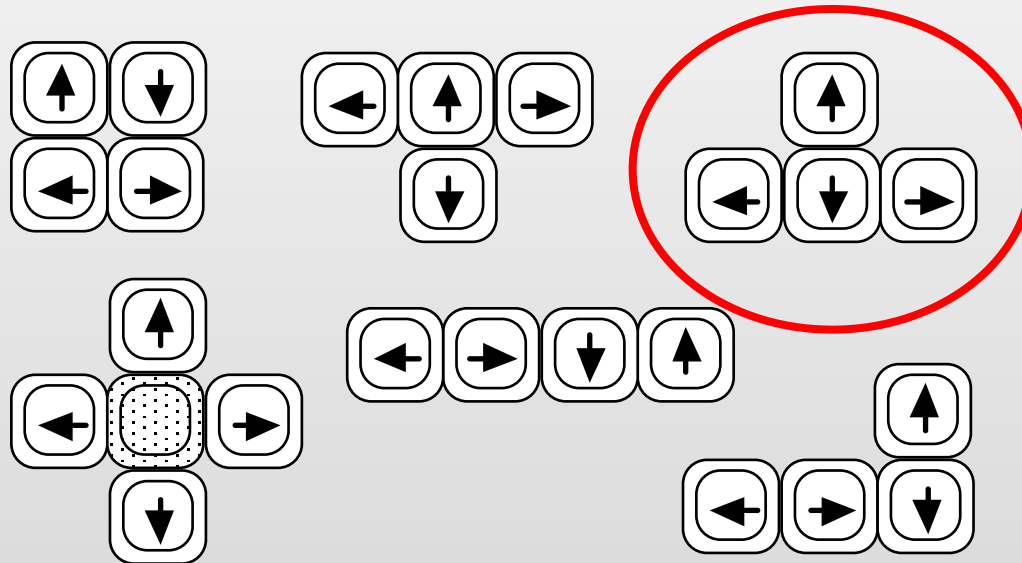
Eyegaze

- control interface by eye gaze direction
- uses laser beam reflected off retina
 - ... a very low power laser!
- mainly used for evaluation
- potential for hands-free control
- high accuracy requires headset
- cheaper and lower accuracy devices available
- sit under the screen like a small webcam



Cursor keys

- Four keys (up, down, left, right) on keyboard.
- Very, very cheap, but slow.
- Useful for not much more than basic motion for text-editing tasks.
- No standardized layout, but inverted “T”, most common



Discrete positioning controls

- in phones, TV controls etc.
 - cursor pads or mini-joysticks
 - discrete left-right, up-down
 - mainly for menu selection

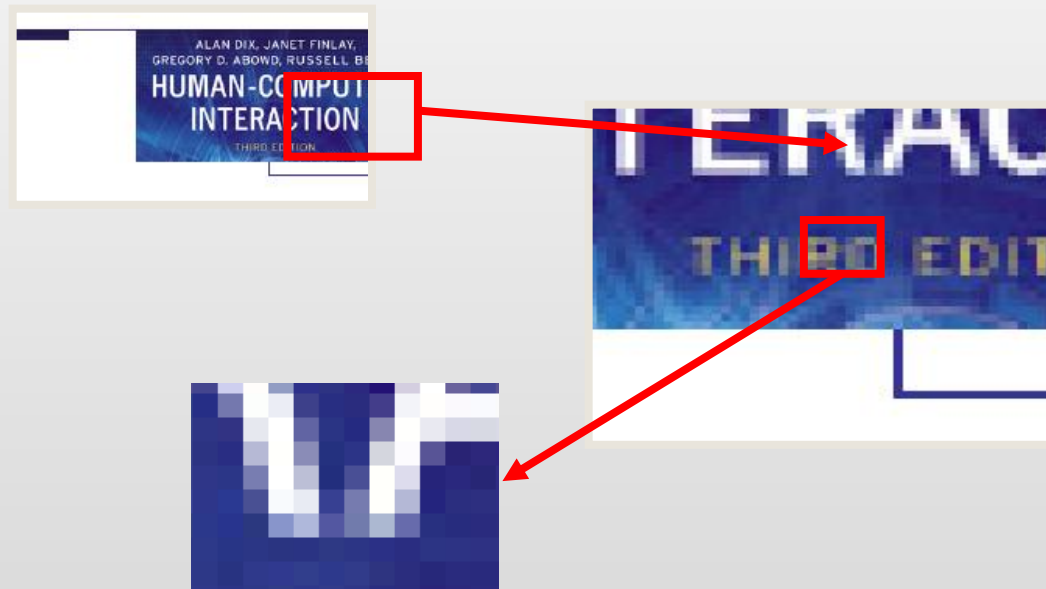




DISPLAY DEVICES

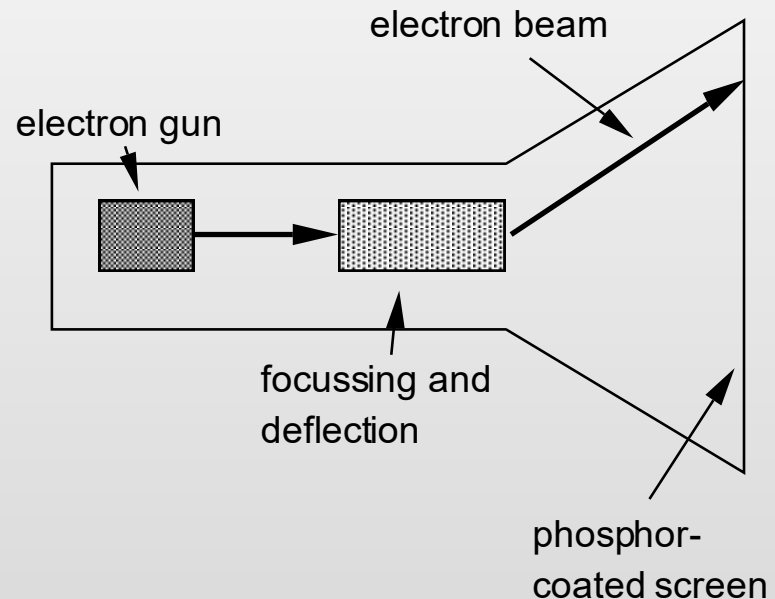
bitmap displays

- made of vast numbers of colored dots or pixels in a rectangular grid.
- These pixels may be limited to black and white (for example, the small display on many TV remote controls), in grayscale, or full color.)



Cathode ray tube

- Stream of electrons emitted from electron gun, focused and directed by magnetic fields, hit phosphor-coated screen which glows
- used in TVs and computer monitors





Liquid crystal displays

- Smaller, lighter, and ... no radiation problems.
- Found on PDAs, portables and notebooks,
... and increasingly on desktop and even for home TV
- also used in dedicated displays:
digital watches, mobile phones,
- How it works ...
 - Top plate transparent and polarised, bottom plate reflecting.
 - Light passes through top plate and crystal, and reflects back to eye.
 - Voltage applied to crystal changes polarisation and hence colour



special displays

Random Scan (Directed-beam refresh, vector display)

- draw the lines to be displayed directly
- no jaggies
- lines need to be constantly redrawn
- rarely used except in special instruments

Direct view storage tube (DVST)

- Similar to random scan but persistent => no flicker
- Can be incrementally updated but not selectively erased
- Used in analogue storage oscilloscopes



large displays

- used for meetings, lectures, etc.

- technology

plasma – usually wide screen

video walls – lots of small screens together

projected – RGB lights or LCD projector

- hand/body obscures screen

- may be solved by 2 projectors + clever software

back-projected

- frosted glass + projector behind



situated displays

- displays in ‘public’ places
 - large or small
 - very public or for small group
- display only
 - for information relevant to location
- or interactive
 - use stylus, touch sensitive screen
- in all cases ... the location matters
 - meaning of information or interaction is related to the location



Example : Hermes a situated display

small displays
beside
office doors



handwritten
notes left
using stylus

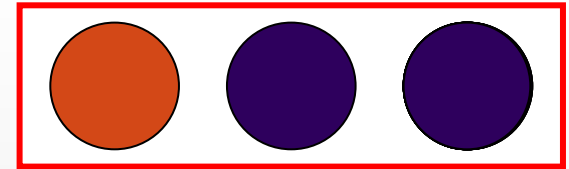


office owner
reads notes
using web interface

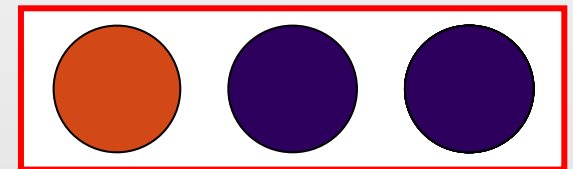
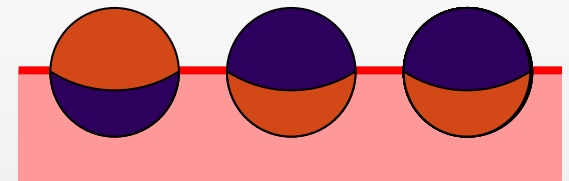
Digital paper

- what?
 - thin flexible sheets
 - updated electronically
 - but retain display
- how?
 - small spheres turned
 - or channels with coloured liquid and contrasting spheres
 - rapidly developing area

appearance



cross section





VIRTUAL REALITY AND 3D INTERACTION

Cockpit controls

- In many PC games and desktop virtual reality (where the output is shown on an ordinary computer screen), the controls are themselves virtual.
- simulated form of the cockpit controls or more prosaic up/down left/right buttons.
- The user manipulates these virtual controls using an ordinary mouse (or other 2D device).



Data glove

- is a 3D input device
- Consists of a lycra glove with optical fibers laid along the fingers, it detects the joint angles of the fingers and thumb.
- As the fingers are bent, the fiber optic cable bends too;
- increasing bend causes more light to leak from the fiber, and the reduction in intensity is detected by the glove and related to the degree of bend in the joint.
- Attached to the top of the glove are two sensors that use ultrasound to determine 3D positional information as well as the angle of roll, that is the degree of wrist rotation



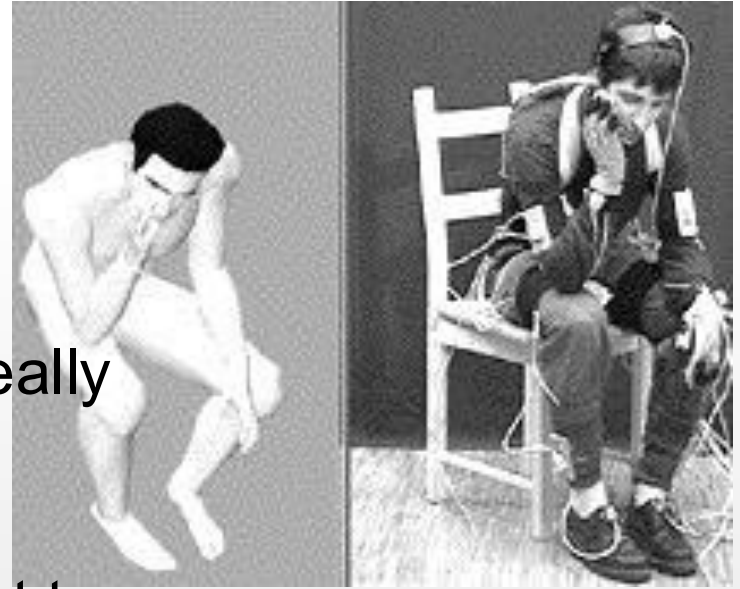
VR Helmet



- The helmets or goggles worn in some VR systems have two purposes:
 - (i) they display the 3D world to each eye and
 - (ii) they allow the user's head position to be tracked
- The head tracking is used primarily to feed into the output side. As the user's head moves around the user ought to see different parts of the scene.
- However, some systems also use the user's head direction to determine the direction of movement within the space and even which objects to manipulate (rather like the eyegaze systems)

Whole body tracking

- make the users feel as if they are really in the virtual world.
- Some VR systems therefore attempt to track different kinds of body movement
- the movement of the whole body may be tracked using devices similar to the dataglove, or using image-processing techniques





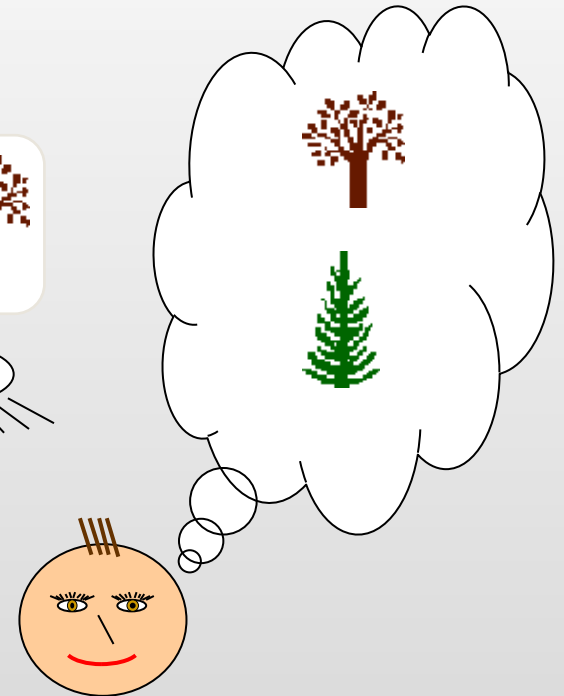
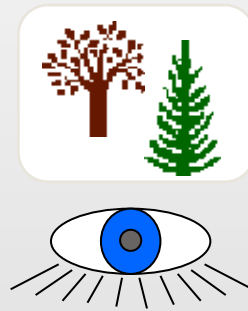
3D displays

1. desktop VR

- ordinary screen, mouse or keyboard control
- perspective and motion give 3D effect
- **seeing in 3D**
 - use stereoscopic vision
 - VR helmets
 - screen plus shuttered specs, etc.

VR headsets

- small TV screen for each eye
- slightly different angles
- 3D effect



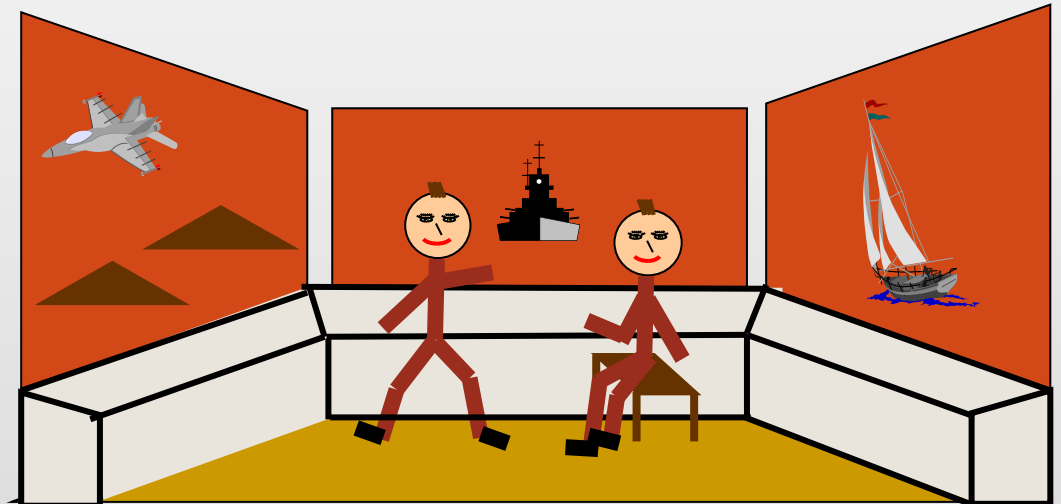
2. VR motion sickness

- time delay
 - move head ... lag ... display moves
 - *conflict*: head movement vs. eyes
- depth perception
 - headset gives different stereo distance
 - but all focused in same plane
 - *conflict*: eye angle vs. focus
- conflicting cues => sickness
 - helps motivate improvements in technology



3. simulators and VR caves

- scenes projected on walls
- realistic environment
- hydraulic rams!
- real controls
- other people





PHYSICAL CONTROLS, SENSORS, ETC



1. Dedicated displays

- analogue representations:
 - dials, gauges, lights, etc.
- digital displays:
 - small LCD screens, LED lights, etc.
- head-up displays
 - found in aircraft cockpits
 - show most important controls
 - ... depending on context



2. Sounds

- beeps, bongs, clonks, whistles and whirrs
- used for error indications
- confirmation of actions e.g. Keyclick



3. Touch, feel, smell

- touch and feeling important
 - in games ... vibration, force feedback
 - in simulation ... feel of surgical instruments called *haptic* devices
- texture, smell, taste
 - current technology very limited



BMW iDrive

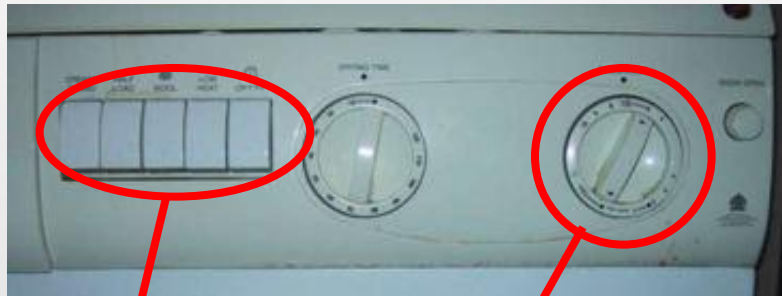
- for controlling menus
- feel small 'bumps' for each item
- makes it easier to select options by feel
- uses haptic technology from Immersion Corp.





4. physical controls

- specialist controls needed ...
 - industrial controls, consumer products, etc.



large buttons

clear dials

tiny buttons



easy-clean
smooth buttons

multi-function
control





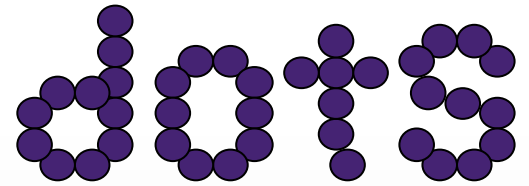
5. Environment and bio-sensing

- sensors all around us
 - car courtesy light – small switch on door
 - ultrasound detectors – security, washbasins
 - RFID security tags in shops
 - temperature, weight, location
- ... and even our own bodies ...
 - iris scanners, body temperature, heart rate, galvanic skin response, blink rate



PAPER PRINTING AND SCANNING

1. Printing



- image made from small dots
 - allows any character set or graphic to be printed,
- critical features:
 - resolution
 - size and spacing of the dots
 - measured in dots per inch (dpi)
 - speed
 - usually measured in pages per minute
 - cost!!



Types of dot-based printers

- dot-matrix printers
 - use inked ribbon (like a typewriter)
 - line of pins that can strike the ribbon, dotting the paper.
 - typical resolution 80-120 dpi
- ink-jet and bubble-jet printers
 - tiny blobs of ink sent from print head to paper
 - typically 300 dpi or better .
- laser printer
 - like photocopier: dots of electrostatic charge deposited on drum, which picks up toner (black powder form of ink) rolled onto paper which is then fixed with heat
 - typically 600 dpi or better.

2. Fonts

- Font – the particular style of text

Courier font

Helvetica font

Palatino font

Times Roman font

- § ' ∞ ≡ ↵ ℞ ☒ ⊗ ↵ ~ ☐ (special symbol)

- Size of a font measured in points (1 pt about 1/72")
(vaguely) related to its height

This is ten point Helvetica

This is twelve point

This is fourteen point

This is eighteen point

and this is twenty-four point

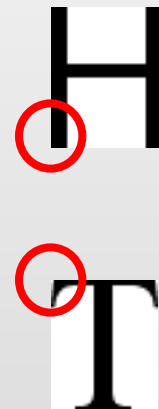
Fonts (ctd)

Pitch

- fixed-pitch – every character has the same width
e.g. `Courier`
- variable-pitched – some characters wider
e.g. Times Roman – compare the ‘i’ and the “m”

Serif or Sans-serif

- sans-serif – square-ended strokes
e.g. Helvetica
- serif – with splayed ends (such as)
e.g. Times Roman or Palatino





Readability of text

- lowercase
 - easy to read shape of words
- UPPERCASE
 - better for individual letters and non-words
e.g. flight numbers: BA793 vs. ba793
- serif fonts
 - helps your eye on long lines of printed text
 - but sans serif often better on screen



3. Page Description Languages

- Pages very complex
 - different fonts, bitmaps, lines, digitised photos, etc.
- Can convert it all into a bitmap and send to the printer
... but often huge !
- PostScript is the most common



4. Screen and page

- WYSIWYG
 - what you see is what you get
 - aim of word processing
- can try to make them similar
but never quite the same
- so ... need different designs, graphics etc, for screen and print



5. Scanners

- Take paper and convert it into a bitmap
- Two sorts of scanner
 - flat-bed: paper placed on a glass plate, whole page converted into bitmap
 - hand-held: scanner passed over paper, digitising strip typically 3-4" wide
- Shines light at paper and note intensity of reflection
 - colour or greyscale
- Typical resolutions from 600–2400 dpi



Scanners (ctd)

Used in

- desktop publishing for incorporating photographs and other images
- document storage and retrieval systems, doing away with paper storage
- + special scanners for slides and photographic negatives



6. Optical character recognition

- The process whereby the computer can ‘read’ the characters on the page.
- It is only comparatively recently that print could be reliably read, since the wide variety of typefaces and print sizes makes this more difficult than one would imagine – it is not simply a matter of matching a character shape to the image on the page.



COMPUTER MEMORY



Short-term Memory - RAM

- Random access memory (RAM)
 - on silicon chips
 - 100 nano-second access time
 - usually volatile (lose information if power turned off)
 - data transferred at around 100 Mbytes/sec
- Some *non-volatile RAM* used to store basic set-up information



Long-term Memory - disks

- magnetic disks
 - floppy disks store around 1.4 Mbytes
 - hard disks typically 40 Gbytes to 100s of Gbytes
access time ~10ms, transfer rate 100kbytes/s
- optical disks
 - use lasers to read and sometimes write
 - more robust than magnetic media
 - CD-ROM
 - same technology as home audio, ~ 600 Mbytes
 - DVD - for AV applications, or very large files

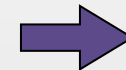


virtual memory

- Problem:
 - running lots of programs + each program large
 - not enough RAM
- Solution - Virtual memory :
 - store some programs temporarily on disk
 - makes RAM appear bigger
- But ... swopping
 - program on disk needs to run again
 - copied from disk to RAM
 - s l o w s t h i n g s d o w n

Compression

- reduce amount of storage required
- lossless
 - recover exact text or image – e.g. GIF, ZIP
 - look for commonalities:
 - text: AAAAAAAAAAABBBBBBCCCCCCCCC 10A5B8C
 - video: compare successive frames and store change
- lossy
 - recover something like original – e.g. JPEG, MP3
 - exploit perception
 - JPEG: lose rapid changes and some colour
 - MP3: reduce accuracy of drowned out notes





Storage formats - text

- ASCII - 7-bit binary code for to each letter and character
- RTF (rich text format)
 - text plus formatting and layout information
- SGML (standardized generalized markup language)
 - documents regarded as structured objects
- XML (extended markup language)
 - simpler version of SGML for web applications



TYPICAL CAPACITIES OF STORAGE MEDIA

	STM small/fast	LTM large/slower
Media:	RAM	Hard disk
Capacity:	256 Mbytes	100 Gbytes
Access time:	10 ns	7 ms
Transfer rate:	100 Mbyte/s	30 Mbyte/s



PROCESSING AND NETWORKS



Finite processing speed

- Designers tend to assume fast processors, and make interfaces more and more complicated
- But problems occur, because processing cannot keep up with all the tasks it needs to do
 - cursor overshooting because system has buffered keypresses
 - icon wars - user clicks on icon, nothing happens, clicks on another, then system responds and windows fly everywhere
- Also problems if system is too fast - e.g. help screens may scroll through text much too rapidly to be read



Limitations on interactive performance

Computation bound

- Computation takes ages, causing frustration for the user

Storage channel bound

- Bottleneck in transference of data from disk to memory

Graphics bound

- Common bottleneck: updating displays requires a lot of effort - sometimes helped by adding a graphics co-processor optimised to take on the burden

Network capacity

- Many computers networked - shared resources and files, access to printers etc. - but interactive performance can be reduced by slow network speed



Networked computing

Networks allow access to ...

- large memory and processing
- other people (groupware, email)
- shared resources – esp. the web

Issues

- network delays – slow feedback
- conflicts - many people update data
- unpredictability



Summary

- A computer system comprises various elements, each of which affects the user of the system.
- Input devices for interactive use, allowing text entry, drawing and selection from the screen:
 - text entry: traditional keyboard, phone text entry, speech and handwriting
 - pointing: principally the mouse, but also touchpad, stylus and others
 - 3D interaction devices.



Summary (cont'd)

- Output display devices for interactive use:
 - different types of screen mostly using some form of bitmap display
 - large displays and situated displays for shared and public use
 - digital paper may be usable in the near future.
- Virtual reality systems and 3D visualization which have special interaction and display devices.



Summary (cont'd)

- Various devices in the physical world:
 - physical controls and dedicated displays
 - sound, smell and haptic feedback
 - sensors for nearly everything including movement, temperature, bio-signs.



Summary (cont'd)

- Various devices in the physical world:
 - physical controls and dedicated displays
 - sound, smell and haptic feedback
 - sensors for nearly everything including movement, temperature, bio-signs.
- Paper output and input: the paperless office and the less-paper office:
 - different types of printers and their characteristics, character styles and fonts
 - scanners and optical character recognition.



Summary (cont'd)

- **Memory:**
 - short-term memory: RAM
 - long-term memory: magnetic and optical disks
 - capacity limitations related to document and video storage
 - access methods as they limit or help the user.
- **Processing:**
 - the effects when systems run too slow or too fast, the myth of the infinitely fast machine
 - limitations on processing speed
 - networks and their impact on system performance.



Activity

- Work on the workbook and answer the exercises 1 to 3 under Activity 4 – Foundation of HCI: The Computer



REFERENCES

- Dix, A. Finlay J., Abowd, G, and Beale, R. (2004) Human Computer Interaction 3rd ed., Pearson Education Limited
- Rogers, Y., Sharp, H and Preece, J. (2007). Interaction Design: Beyond Human Computer Interaction, Second Edition, Wiley & Sons
- Norman, D. (2013). The DESIGN of EVERYDAY THINGS. Basic Books, A Member of the Perseus Books Group